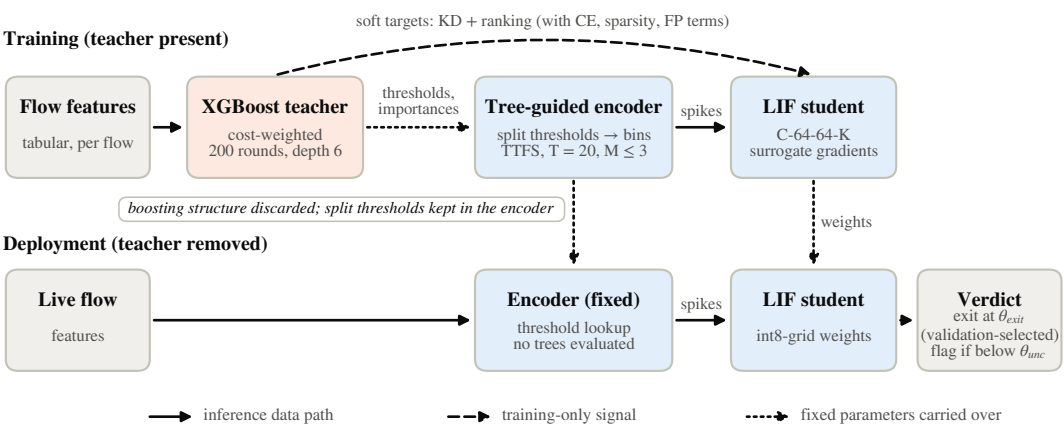
soft targets: KD + ranking (with CE, sparsity, FP terms)
Training (teacher present)
Flow features
tabular, per flow
XGBoost teacher
cost-weighted
200 rounds, depth 6
thresholds, importances
Tree-guided encoder
split thresholds → bins
TTFS, T = 20, M ≤ 3
spikes
LIF student
C-64-64-K
surrogate gradients
boosting structure discarded; split thresholds kept in the encoder
weights
Deployment (teacher removed)
Live flow
features
Encoder (fixed)
threshold lookup
no trees evaluated
spikes
LIF student
int8-grid weights
Verdict
exit at θ_{exit}
(validation-selected)
flag if below θ_{unc}
inference data path
training-only signal
fixed parameters carried over